

RESEARCH ARTICLE

WILEY Heteroatom Chemistry

Synthesis of 4-aryl- and 4-acyl-1,3-dithiole-2-thiones via deprotonative zincation of 1,3-dithiole-2-thione

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Abstract

We have developed deprotonative zincation of 1,3-dithiole-2-thione at the 4-position. Addition of lithium diisopropylamide to a THF solution of 1,3-dithiole-2-thione containing ZnI_2 and LiI generated the corresponding organozinc species. After this zincation, various aryl and acyl groups were installed at the 4-position by Pd-catalyzed cross-coupling reaction with aryl iodides and Cu-catalyzed acylation with acyl chlorides, respectively.

1 | INTRODUCTION

Tetrathiafulvalenes (TTFs)^[1] have received much attention in structural organic and material chemistry due to the characteristic redox behavior of the 1,3-dithiole rings, which can gain aromatic stabilization by one-electron oxidation to the corresponding cationic species. To tune the electronic property of TTFs, various methods to introduce substituents have been developed, including deprotonative metalation^[2] and our direct mono-, di-, or tetraarylation.^[3] Recently, π -extended TTFs, which contain a 1,3-dithiole ring attached to a π system through an exo double bond at the 2-position, are also receiving increasing attention.^[4] As 1,3-dithiole-2-thione (**1**) is a precursor of not only TTFs but also π -extended TTFs, development of efficient and reliable synthetic routes to substituted 1,3-dithiole-2-thiones would definitely contribute to the development of TTF chemistry.

The most general synthetic method of 4-substituted 1,3-dithiole-2-thiones is functionalization of parent **1**. Halo,^[5] stannyl,^[6] or aryl(hydroxy)methyl^[7] groups can be introduced by deprotonation of **1** with *n*BuLi or lithium diisopropylamide (LDA) followed by trapping with electrophiles (Scheme 1A). Very recently, Knochel and co-worker developed deprotonative magnesiation of **1** with $\text{TMPMgCl}\cdot\text{LiCl}$ (TMP = 2,2,6,6-tetramethylpiperidyl) (Scheme 1B).^[8] Although this method

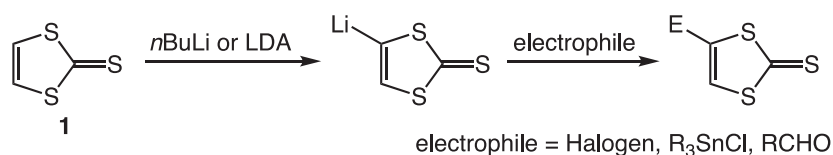
is applicable to introducing a variety of substituents, transmetalation of the resulting organomagnesium species with ZnCl_2 is necessary to accomplish subsequent arylation or acylation reactions. Here, we report a straightforward deprotonative zincation of **1** by means of LDA in the presence of ZnI_2 and LiI and one-pot Negishi-type cross-coupling with aryl iodides or acyl chlorides (Scheme 1C).

2 | RESULTS AND DISCUSSION

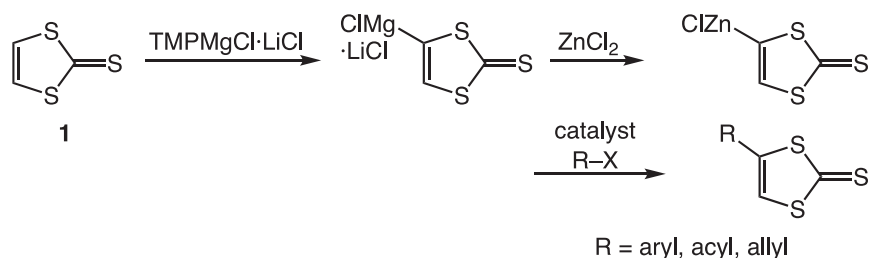
$\text{TMPZnCl}\cdot\text{LiCl}$,^[9] which is widely used for deprotonative zincation, was at first tried to deprotonate 1,3-dithiole-2-thione at the 4-position. However, the addition of $\text{TMPZnCl}\cdot\text{LiCl}$ to 1,3-dithiole-2-thione at 0°C followed by an addition of iodine did not afford 4-iodo-1,3-dithiole-2-thione (**2**). This result indicates that deprotonation with $\text{TMPZnCl}\cdot\text{LiCl}$ did not proceed. We then examined the possibility of deprotonation with a strongly basic lithium amide in the presence of ZnCl_2 and LiCl to achieve zincation.^[10] Specifically, the addition of LiTMP in the presence of ZnCl_2 and LiCl followed by the addition of iodine afforded the mono-iodinated product **2** (Table 1). Among various lithium bases, LDA showed the best result (entry 5). The addition of 1.5 equiv of LDA in the presence of 1.1 equiv of ZnCl_2 and 2.2 equiv of LiCl followed by trapping with iodine afforded **2** in 87% yield.

Dedicated to Professor Naomichi Furukawa on the occasion of his 82nd birthday

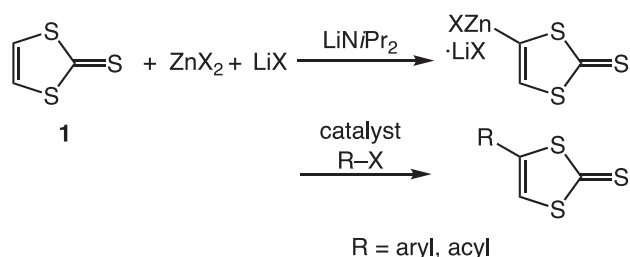
(A) Deprotonation with lithium base followed by trapping with electrophiles



(B) Deprotonation with $TMPMgCl \cdot LiCl$, transmetalation to $ZnCl_2$, and cross-coupling



(C) **This Work**



SCHEME 1 Methods to synthesize substituted 1,3-dithiole-2-thiones

TABLE 1 Optimization of deprotonative zincation

Entry	Base	NMR yield (%)
1	LiTMP	78
2	LiOtBu	1
3	<i>n</i> BuLi	58
4	$LiN(SiMe_3)_2$	40
5	$LiNiPr_2$	87

Note. Conditions: **1** (0.50 mmol), $ZnCl_2$ (1.1 equiv), $LiCl$ (2.2 equiv), base (1.5 equiv), THF (1.0 mL), then I_2 (2.0 equiv).

With the optimal conditions for deprotonative zincation in hand, we attempted Negishi-type cross-coupling of the organozinc species. A variety of palladium catalysts

(Figure 1) were examined for the model reaction with *p*-iodotoluene (Table 2, entries 1–8). As a result, bisphosphine-ligated Pd complexes generally showed better catalytic

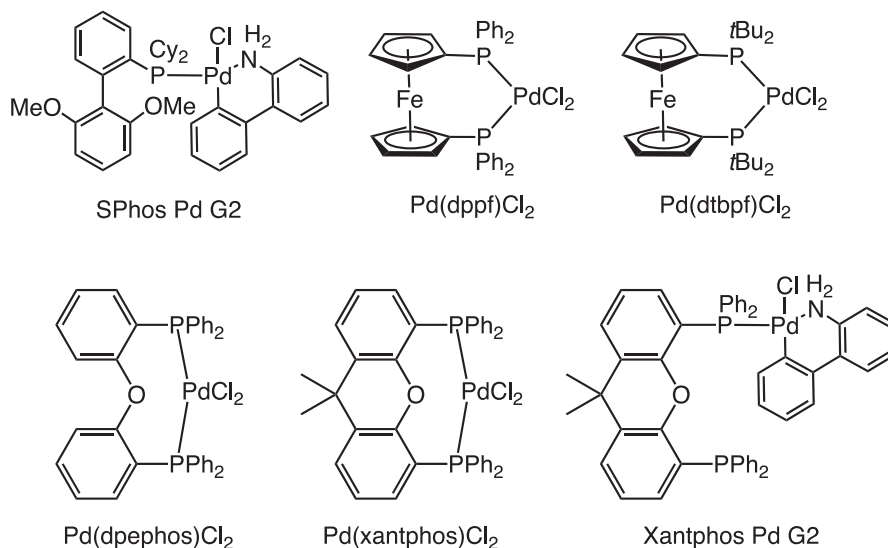


FIGURE 1 Structures of catalysts used in this work

TABLE 2 Optimization of arylation

Entry	Catalyst	X	X'	Temp. (°C)	NMR yield (%)
1	SPhos Pd G2	Cl	Cl	80	<1
2	Pd(PPh ₃) ₂ Cl ₂	Cl	Cl	80	<1
3	Pd[P(2-furyl) ₃] ₂ Cl ₂	Cl	Cl	80	<1
4	Pd[P(4-F-C ₆ H ₄) ₃] ₂ Cl ₂	Cl	Cl	80	12
5	Pd(dppf)Cl ₂	Cl	Cl	80	25
6	Pd(dtbpf)Cl ₂	Cl	Cl	80	22
7	Pd(dpephos)Cl ₂	Cl	Cl	80	47
8	Pd(xantphos)Cl ₂	Cl	Cl	80	65
9	Pd(xantphos)Cl ₂	I	I	25	77
10	Pd(xantphos)Cl ₂	Cl	Cl	25	13
11	Pd(xantphos)Cl ₂	I	Cl	25	<1
12	Pd(xantphos)Cl ₂	Cl	I	25	<1
13	Xantphos Pd G2	I	I	25	82
14	Xantphos Pd G2	I	I	35	96 ^a

Note. Conditions: **1** (0.50 mmol), ZnX₂ (1.1 equiv), LiX' (2.2 equiv), LDA (2.0 M in THF/heptane/ethylbenzene, 1.5 equiv), THF (1.0 mL), then catalyst (5 mol%), *p*-iodotoluene (1.5 equiv), 12 hr.

^aReaction time: 3 hr.

activity than monophosphine-ligated ones,^a and above all, a Pd/Xantphos system provided monoarylated product **3a** in 65% yield (entry 8). After further screening of reaction conditions, this reaction was found to proceed at 25°C using

ZnI₂ and LiI instead of ZnCl₂ and LiCl, providing **3a** in 77% yield (entry 9).^b Both ZnI₂ and LiI were crucial for the reaction at 25°C (entries 10–12). Employment of Xantphos Pd G2 precatalyst^[12] instead of Pd(xantphos)Cl₂ increased the

^aWe assume that the difference between monodentate ligands and bidentate ones lies in the robustness of the palladium–phosphine complexes.

^bIt was reported that PhZnI–LiI is more reactive than PhZnCl–LiCl in transition metal-catalyzed reaction. See:^[11]

TABLE 3 Scope of arylation

Entry	Ar	Time	Product	Yield (%)
1	4-MeC ₆ H ₄	3 hr	3a	90
2	4-CF ₃ C ₆ H ₄	3 hr	3b	79
3	4-MeOC ₆ H ₄	3 hr	3c	76
4	4-NCC ₆ H ₄	4 hr	3d	87
5	4-MeOCOC ₆ H ₄	2 hr	3e	97 ^a
6	4-HCOC ₆ H ₄	6 hr	3f	68 ^a
7	4-PhC ₆ H ₄	4 hr	3g	66
8	3-thienyl	3 hr	3h	81

Notes. Conditions: **1** (0.50 mmol), ZnI₂ (1.1 equiv), LiI (2.2 equiv), LDA (2.0 M in THF/heptane/ethylbenzene, 1.5 equiv), THF (1.0 mL), then Xantphos Pd G2 (5 mol%), Ar-I (1.5 equiv).

^aBefore arylation, volatiles were removed under reduced pressure and THF (1.5 mL) was added.

TABLE 4 Optimization of acylation

Entry	X	NMR yield (%)
1	I	0
2	I	64 ^a
3	Cl	56 ^a
4	I	63 ^{a,b}

Notes. Conditions: **1** (0.50 mmol), ZnX₂ (1.1 equiv), LiX (2.2 equiv), LDA (2.0 M in THF/heptane/ethylbenzene, 1.2 equiv), THF (1.0 mL), then CuI (10 mol%), pivaloyl chloride (2.0 equiv).

^aBefore acylation, volatiles were removed under reduced pressure and THF (1.5 mL) was added.

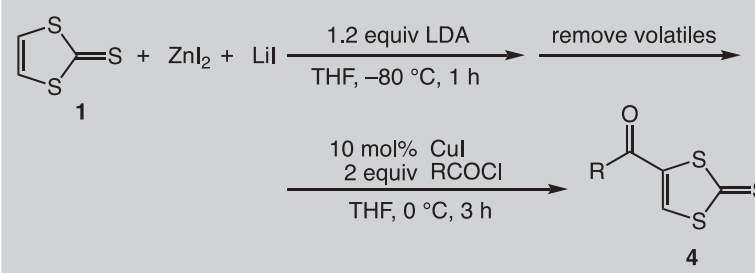
^bCuCN was used instead of CuI.

yield (entry 13) and the reaction at 35 °C afforded a 96% yield of **3a** in 3 hr (entry 14).

Thanks to the mild reactivity of the corresponding organozinc species, aryl iodide could contain various functional groups. Neither electron-withdrawing trifluoromethyl nor electron-donating methoxy groups retarded the reaction (Table 3, entries 2 and 3). 4-Iodobenzonitrile reacted smoothly to provide **3d** in excellent yield without loss of the cyano group (entry 4). Employment of aryl iodide containing an ester or formyl group resulted in a poor yield of the

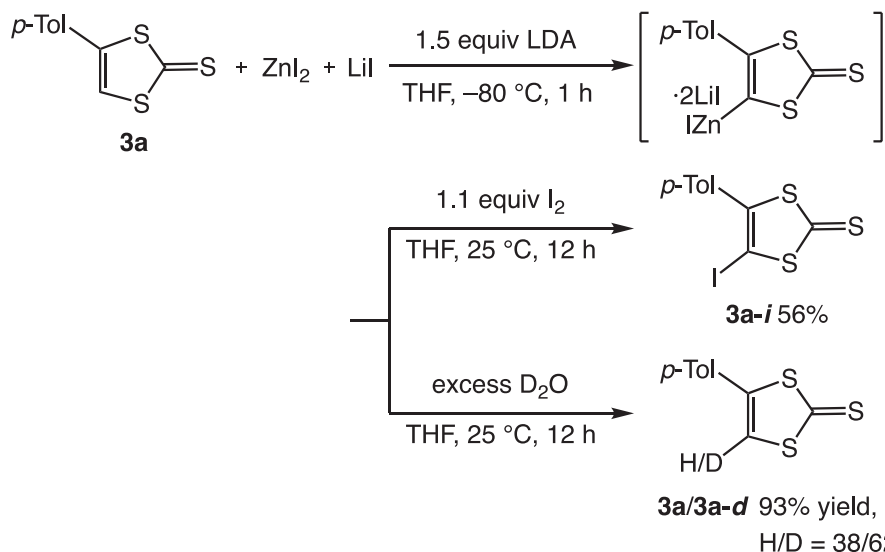
desired product under the standard conditions. Fortunately, when the mixtures were once evacuated to remove diisopropylamine before arylation, the yields of **3e** and **3f** increased to 97% and 68%, respectively (entries 5 and 6). π -Extended 4-iodobiphenyl and heteroaromatic 3-iodothiophene could participate in this reaction to afford 4-(1,1'-biphenyl-4-yl)- and 4-(3-thienyl)-1,3-dithiole-2-thione (**3g** and **3h**) in high yields (entries 7 and 8).

As we succeeded in arylation, we then attempted acylation of the organozinc species with acyl chlorides. However, the

TABLE 5 Scope of acylation


Entry	R	Product	Yield (%)
1	<i>t</i> Bu	4a	61
2	4-O ₂ NC ₆ H ₄	4b	70
3	4-MeOC ₆ H ₄	4c	52
4	CH ₂ =CH	4d	52
5	2-thienyl	4e	64
6	3-thienyl	4f	56

Notes. Conditions: **1** (0.50 mmol), ZnI₂ (1.1 equiv), LiI (2.2 equiv), LDA (2.0 M in THF/heptane/ethylbenzene, 1.2 equiv), THF (1.0 mL), then CuI (10 mol%), acyl chloride (2.0 equiv), THF (1.5 mL).

**SCHEME 2** Deprotonative zincation of **3a** with LDA in the presence of ZnI₂ and LiI

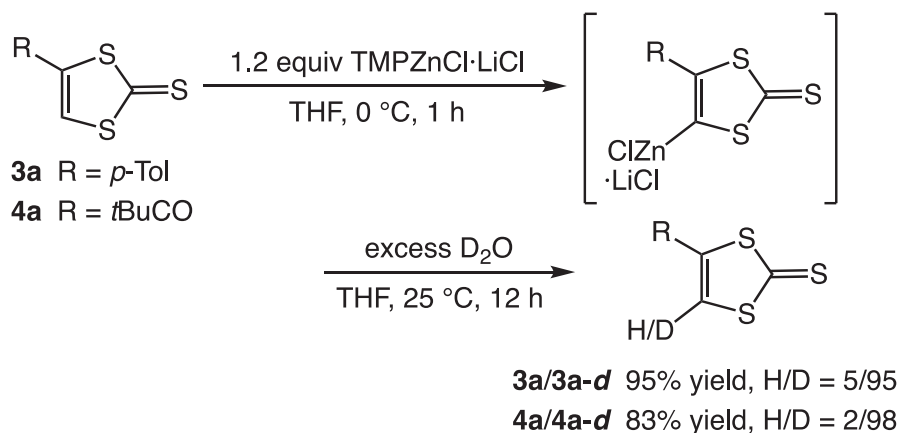
addition of 10 mol% of CuI and 2.0 equiv of pivaloyl chloride to the organozinc solution did not afford 4-pivaloyl-1,3-dithiophene-2-thione (**4a**) at all (Table 4, entry 1). This is probably because diisopropylamine reacted with pivaloyl chloride. In fact, removal of volatiles including diisopropylamine and addition of THF as a solvent before acylation changed the result dramatically to provide the desired product **4a** in 64% yield (entry 2). Also in this case, iodide counteranion was more suitable than chloride (entry 3). CuCN also catalyzed the reaction similarly (entry 4).

Various acyl groups could be introduced with the method described in Table 4, entry 2. Aryl groups containing electron-withdrawing nitro or electron-donating methoxy groups were successfully installed (Table 5, entries 2 and 3). Installation of an acryloyl group was also possible in moderate yield (entry 4). Heteroaromatic

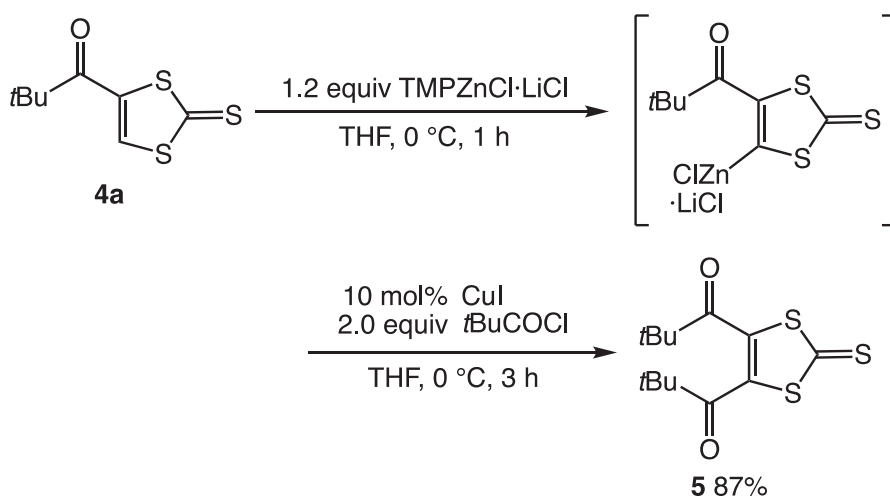
thiophenecarbonyl groups were also introduced (entries 5 and 6).

Next, we tried to metalate the 5-position of 4-aryl- and 4-acyl-1,3-dithiophene-2-thiones. Unfortunately, the same method showed lower efficiency. Deprotonative zincation of **3a** through the above-described method followed by treatment with I₂ or D₂O afforded a 56% yield of **3a-i** or 58% of **3a-d**, respectively (Scheme 2), which indicates that deprotonative zincation was inefficient.^c Surprisingly, however, we found that TMPZnCl-LiCl was an exquisite base to achieve the desired metalation; deprotonative zincation of **3a** and **4a** followed by additions of D₂O provided **3a-d** and **4a-d**,

^cWhen LiTMP was used instead of LDA, the yield of the deuterated compound increased to 86%.



SCHEME 3 Deprotonative zincation of 4-substituted 1,3-dithiole-2-thiones with TMPZnCl·LiCl



SCHEME 4 Acylation of 4-pivaloyl-1,3-dithiole-2-thione

respectively, in high yields and with high selectivities without touching the acyl group on **4a** (Scheme 3). The organozinc reagent prepared from **4a** with TMPZnCl·LiCl was also applicable to copper-catalyzed acylation with pivaloyl chloride to provide 4,5-diacyl-1,3-dithiole-2-thione **5** (Scheme 4).

3 | CONCLUSION

Deprotonative zincation of 1,3-dithiole-2-thione with lithium diisopropylamide in the presence of zinc iodide and lithium iodide was developed. The organozinc species prepared with this method could be applied to Negishi coupling with aryl iodides and acyl chlorides to efficiently provide 4-aryl- and 4-acyl-1,3-dithiole-2-thione, respectively. The use of TMPZnCl·LiCl enabled deprotonation of the 5-position of 4-aryl- and 4-acyl-1,3-dithiole-2-thiones, allowing 4,5-difunctionalization.

4 | EXPERIMENTAL SECTION

4.1 | Instrumentation and Chemicals

¹H NMR (600 MHz) and ¹³C NMR (151 MHz) spectra were taken on a JEOL ECA-600 spectrometer and were recorded

in CDCl₃. Chemical shifts (δ) are in parts per million relative to chloroform at 7.26 ppm for ¹H and relative to CDCl₃ at 77.16 ppm for ¹³C. TLC analyses were performed on commercial aluminum plates bearing a 0.25-mm layer of Merck Silica gel 60F₂₅₄. Silica gel (Wakosil C-300) was used for column chromatography. Mass spectra were taken on a Bruker micrOTOF II-KR spectrometer. Unless otherwise noted, materials obtained from commercial suppliers were used without further purification. 1,3-Dithiole-2-thione was purchased from TCI. Aryl iodides and acyl chlorides were purchased from Aldrich, TCI, or Wako Pure Chemical Industries, Ltd. 2,2,6,6-Tetramethylpiperidine were purchased from Wako Pure Chemical Industries, Ltd. Zinc chloride (265-00275), zinc iodide (125-03322), lithium chloride (265-00312), and lithium iodide (122-03452) were purchased from Wako Pure Chemical Industries, Ltd., dried under reduced pressure at 160 °C for 3 hr, and stored in a glove box filled with argon. Dehydrated THF was purchased from KANTO CHEMICAL CO., INC. and stored under nitrogen with Grubbs apparatus.

4.1.1 | 4-Iodo-1,3-dithiole-2-thione (2)

Zinc chloride (75.0 mg, 0.55 mmol), lithium chloride (46.6 mg, 1.1 mmol), and 1,3-dithiole-2-thione (67.1 mg,

0.50 mmol) were dissolved in THF (1.0 mL). After the solution was cooled to -80°C , lithium base (0.75 mmol) was added dropwise. The resulting mixture was stirred at the temperature for 1 hr and warmed up to 25°C . After the addition of iodine (254 mg, 1.0 mmol), the resulting mixture was stirred at room temperature for 1 hr. The mixture was diluted with a mixture of hexane and EtOAc (5/1, 5 mL) and passed through a pad of silica gel with the same solvent mixture. The crude mixture was analyzed by ^1H NMR by using dibenzyl ether as an internal standard.

4.1.2 | 4-Aryl-1,3-dithiole-2-thione 3; general procedure 1 (GP1)

Zinc iodide (175.6 mg, 0.55 mmol), lithium iodide (147.2 mg, 1.1 mmol), and 1,3-dithiole-2-thione (67.1 mg, 0.50 mmol) were dissolved in THF (1.0 mL). After the solution was cooled to -80°C , lithium diisopropylamide (2.0 M in THF/heptane/ethylbenzene, 0.375 mL, 0.75 mmol) was added dropwise. The resulting mixture was stirred at the temperature for 1 hr and warmed to 25°C . After additions of Xantphos Pd G2 (22.2 mg, 0.025 mmol) and aryl iodide (0.75 mmol), the resulting mixture was stirred at 35°C for indicated reaction time and quenched with water (3 mL). The mixture was poured into sat. NaHCO_3 aq. (40 mL) and organic compounds were extracted with CH_2Cl_2 (10 mL $\times$ 3). The combined organic layer was dried over anhydrous sodium sulfate, passed through a pad of activated alumina, and concentrated in vacuo. Silica gel column chromatography (hexane/ CH_2Cl_2 = 2/1) provided **3**.

4.1.3 | 4-(4-Methylphenyl)-1,3-dithiole-2-thione (3a)

According to GP1, **3a** (101.4 mg, 90%) was obtained as a yellow solid. ^1H NMR δ = 7.31 (d, J = 8.1 Hz, 2H), 7.23 (d, J = 8.1 Hz, 2H), 7.05 (s, 1H), 2.39 (s, 3H) ppm; ^{13}C NMR δ = 212.6, 146.7, 140.2, 130.2, 128.5, 126.5, 121.1, 21.4 ppm; HR-MS (APCI-MS, positive): m/z = 224.9870. Calcd for $\text{C}_{10}\text{H}_9\text{S}_3^+$: 224.9861 [$M+\text{H}$] $^+$.

4.1.4 | 4-(4-Trifluoromethylphenyl)-1,3-dithiole-2-thione (3b)

According to GP1, **3b** (109.6 mg, 79%) was obtained as a yellow solid. ^1H NMR δ = 7.54 (d, J = 7.0 Hz, 2 H), 7.70 (d, J = 7.0 Hz, 2H), 7.23 (s, 1H) ppm; ^{13}C NMR δ = 211.6, 144.2, 134.4, 131.6 (q, J = 33.0 Hz), 126.9, 126.6, 126.6, 123.7 (q, J = 273 Hz) ppm; HR-MS (APCI-MS, positive): m/z = 278.9586. Calcd for $\text{C}_{10}\text{H}_6\text{F}_3\text{S}_3^+$: 278.9578 [$M+\text{H}$] $^+$.

4.1.5 | 4-(4-Methoxyphenyl)-1,3-dithiole-2-thione (3c)

According to GP1, **3c** (91.0 mg, 76%) was obtained as a yellow solid. ^1H NMR δ = 7.35 (d, J = 8.9 Hz, 2H), 6.96 (s, 1H), 6.94 (d, J = 8.9 Hz, 2H), 3.84 (s, 3H); ^{13}C NMR δ = 212.6, 160.9, 146.4, 128.1, 123.9, 120.1, 114.9, 55.6; HR-MS (APCI-MS, positive): m/z = 240.9812. Calcd for $\text{C}_{10}\text{H}_9\text{S}_3\text{O}^+$: 240.9810 [$M+\text{H}$] $^+$.

4.1.6 | 4-(4-Cyanophenyl)-1,3-dithiole-2-thione (3d)

According to GP1, **3d** (97.7 mg, 83%) was obtained as a yellow solid. ^1H NMR δ = 7.73 (d, J = 8.9 Hz, 2H), 7.53 (d, J = 8.9 Hz, 2H), 7.28 (s, 1H) ppm; ^{13}C NMR δ = 211.0, 143.5, 135.1, 133.3, 127.1, 124.9, 118.0, 113.3 ppm; HR-MS: m/z = 235.9663. Calcd for $\text{C}_{10}\text{H}_6\text{NS}_3^+$: 235.9657 [$M+\text{H}$] $^+$.

4.1.7 | 4-(4-Methoxycarbonylphenyl)-1,3-dithiole-2-thione (3e)

Zinc iodide (175.6 mg, 0.55 mmol), lithium iodide (147.2 mg, 1.1 mmol), and 1,3-dithiole-2-thione (67.1 mg, 0.50 mmol) were dissolved in THF (1.0 mL). After the solution was cooled to -80°C , lithium diisopropylamide (2.0 M in THF/heptane/ethylbenzene, 0.375 mL, 0.75 mmol) was added dropwise. The resulting mixture was stirred at the temperature for 1 hr and warmed to 25°C . All volatiles were removed under reduced pressure and fresh THF (1.5 mL) was added. After additions of Xantphos Pd G2 (22.2 mg, 0.025 mmol) and methyl 4-iodobenzoate (197 mg, 0.75 mmol), the resulting mixture was stirred at 35°C for 2 hr and quenched with water (3 mL). A solid formed was collected by filtration and washed with water and ethanol (10 mL each) to provide **3e** (130.5 mg, 97%) as a yellow solid. ^1H NMR δ = 8.09 (d, J = 8.1 Hz, 2H), 7.49 (d, J = 8.1 Hz, 2H), 7.24 (s, 1H), 3.95 (s, 3H) ppm; ^{13}C NMR δ = 211.7, 166.2, 144.9, 135.1, 131.1, 130.8, 126.5, 123.9, 52.6 ppm; HR-MS (APCI-MS, positive): m/z = 268.9764. Calcd for $\text{C}_{11}\text{H}_9\text{O}_2\text{S}_3^+$: 268.9759 [$M+\text{H}$] $^+$.

4.1.8 | 4-(4-Formylphenyl)-1,3-dithiole-2-thione (3f)

Zinc iodide (175.6 mg, 0.55 mmol), lithium iodide (147.2 mg, 1.1 mmol), and 1,3-dithiole-2-thione (67.1 mg, 0.50 mmol) were dissolved in THF (1.0 mL). After the solution was cooled to -80°C , lithium diisopropylamide (2.0 M in THF/heptane/ethylbenzene, 0.375 mL, 0.75 mmol) was added dropwise. The resulting mixture was stirred at the temperature for 1 hr and warmed to 25°C . All volatiles were removed under reduced pressure and fresh THF (1.5 mL) was added.

After additions of Xantphos Pd G2 (22.2 mg, 0.025 mmol) and 4-iodobenzaldehyde (174 mg, 0.75 mmol), the resulting mixture was stirred at 35°C for 6 hr and quenched with water (3 mL). The mixture was poured into sat. NaHCO₃ aq. (40 mL) and organic compounds were extracted with CH₂Cl₂ (10 mL×3). The combined organic layer was dried over anhydrous sodium sulfate, passed through a pad of activated alumina, and concentrated in vacuo. Silica gel column chromatography (hexane/CH₂Cl₂ = 2/1) provided **3f** (80.9 mg, 68%) as a yellow solid. ¹H NMR δ = 10.05 (s, 1H), 7.95 (d, *J* = 8.2 Hz, 2H), 7.59 (d, *J* = 8.2 Hz, 2H), 7.29 (s, 1H) ppm; ¹³C NMR δ = 211.4, 191.1, 144.4, 136.8, 136.3, 130.8, 127.1, 124.6 ppm; HR-MS (APCI-MS, positive): *m/z* = 238.9650. Calcd for C₁₀H₇OS₃⁺: 238.9654 [*M*+H]⁺.

4.1.9 | 4-(1,1'-Biphenyl-4-yl)-1,3-dithiole-2-thione (3g)

According to GP1, **3g** (94.2 mg, 66%) was obtained as a yellow solid. ¹H NMR δ = 7.66 (d, *J* = 8.1 Hz, 2H), 7.60 (d, *J* = 7.5 Hz, 2H), 7.50 (d, *J* = 8.1 Hz, 2H), 7.47 (t, *J* = 7.5 Hz, 2H), 7.39 (t, *J* = 7.5 Hz, 1H), 7.15 (s, 1H) ppm; ¹³C NMR δ = 212.3, 146.1, 142.7, 139.9, 130.0, 129.1, 128.2, 128.1, 127.2, 127.0, 121.8 ppm; HR-MS: *m/z* = 287.0018. Calcd for C₁₅H₁₁S₃⁺: 287.0017 [*M*+H]⁺.

4.1.10 | 4-(3-Thienyl)-1,3-dithiole-2-thione (3h)

According to GP1, **3h** (87.4 mg, 81%) was obtained as a yellow solid. ¹H NMR δ = 7.41 (dd, *J* = 4.8, 2.7 Hz, 1H), 7.34 (dd, *J* = 2.7, 1.4 Hz, 1H), 7.20 (dd, *J* = 4.8, 1.4 Hz, 1H), 7.04 (s, 1H) ppm; ¹³C NMR δ = 212.0, 141.0, 132.0, 127.8, 125.3, 123.7, 121.4 ppm; HR-MS (APCI-MS, positive): *m/z* = 216.9268. Calcd for C₇H₅S₄: 216.9269 [*M*+H]⁺.

4.1.11 | 4-Acyl-1,3-dithiole-2-thione 4; General Procedure 2 (GP2)

Zinc iodide (175.6 mg, 0.55 mmol), lithium iodide (147.2 mg, 1.1 mmol), and 1,3-dithiole-2-thione (67.1 mg, 0.50 mmol) were dissolved in THF (1.0 mL). After the solution was cooled to -80°C, lithium diisopropylamide (2.0 M in THF/heptane/ethylbenzene, 0.30 mL, 0.60 mmol) was added dropwise. The resulting mixture was stirred at the temperature for 1 hr and warmed to 25°C. All volatiles were removed under reduced pressure and THF (1.5 mL) was added. After additions of CuI (9.5 mg, 0.050 mmol) and acyl chloride (1.0 mmol) at 0°C, the resulting mixture was stirred at the same temperature for 3 hr and quenched with water (3 mL). The mixture was poured into sat. NaHCO₃ aq. (40 mL)

and organic compounds were extracted with CH₂Cl₂ (10 mL×3). The combined organic layer was dried over anhydrous sodium sulfate, passed through a pad of activated alumina, and concentrated in vacuo. Silica gel column chromatography (hexane/CH₂Cl₂ = 2/1) provided **4**.

4.1.12 | 4-Pivaloyl-1,3-dithiole-2-thione (4a)

According to GP2 starting with 3 mmol of **1**, **4a** (422 mg, 64%) was obtained as a yellow solid. ¹H NMR δ = 7.86 (s, 1H), 1.34 (s, 9H) ppm; ¹³C NMR δ = 211.2, 195.1, 144.3, 136.2, 44.2, 27.5 ppm; HR-MS (APCI-MS, positive): *m/z* = 218.9965. Calcd for C₈H₁₁OS₃⁺: 218.9967 [*M*+H]⁺.

4.1.13 | 4-(4-Nitrobenzoyl)-1,3-dithiole-2-thione (4b)

According to GP2, **4b** (98.9 mg, 70%) was obtained as a brown solid. ¹H NMR δ = 8.39 (d, *J* = 8.9 Hz, 2H), 7.96 (d, *J* = 8.9 Hz, 2H), 7.73 (s, 1H) ppm; ¹³C NMR δ = 210.1, 182.3, 150.6, 145.1, 140.9, 140.5, 130.0, 124.3 ppm; HR-MS (APCI-MS, positive): *m/z* = 283.9518. Calcd for C₁₀H₆NO₂S₃⁺: 283.9504 [*M*+H]⁺.

4.1.14 | 4-(4-Methoxybenzoyl)-1,3-dithiole-2-thione (4c)

According to GP2, **4c** (69.7 mg, 52%) was obtained as a yellow solid. ¹H NMR δ = 7.83 (d, *J* = 8.8 Hz, 2H), 7.67 (s, 1H), 7.00 (d, *J* = 8.8 Hz, 2H), 3.90 (s, 3H) ppm; ¹³C NMR δ = 211.5, 182.6, 164.3, 146.4, 137.3, 131.7, 128.4, 114.5, 55.8 ppm; HR-MS (APCI-MS, positive): *m/z* = 268.9750. Calcd for C₁₁H₉O₂S₃⁺: 268.9759 [*M*+H]⁺.

4.1.15 | 4-Acryloyl-1,3-dithiole-2-thione (4d)

According to GP2, **4d** (49.3 mg, 52%) was obtained as a yellow solid. ¹H NMR δ = 7.87 (s, 1H), 6.85 (dd, *J* = 17.0, 10.9 Hz, 1H), 6.53 (d, *J* = 17.0 Hz, 1H), 6.01 (d, *J* = 10.9 Hz, 1H) ppm; ¹³C NMR δ = 211.1, 178.7, 147.3, 136.9, 132.4, 129.5 ppm; HR-MS (APCI-MS, positive): *m/z* = 188.9491. Calcd for C₆H₅O₂S₃⁺: 188.9497 [*M*+H]⁺.

4.1.16 | 4-(2-Thiophenecarbonyl)-1,3-dithiole-2-thione (4e)

According to GP2, **4e** (77.7 mg, 64%) was obtained as a yellow solid. ¹H NMR δ = 7.94 (s, 1H), 7.84 (m, 1H), 7.80 (d, *J* = 4.8 Hz, 1H), 7.22 (m, 1H) ppm; ¹³C NMR δ = 210.9, 174.7, 145.3, 140.7, 136.8, 135.6, 134.1, 128.7 ppm; HR-MS (APCI-MS, positive): *m/z* = 244.9218. Calcd for C₈H₅OS₄⁺: 244.9218 [*M*+H]⁺.

4.1.17 | 4-(3-Thiophenecarbonyl)-1,3-dithiole-2-thione (4f)

According to GP2, **4f** (65.7 mg, 56%) was obtained as a yellow solid. ^1H NMR δ = 8.08 (dd, J = 3.4, 1.4 Hz, 1H), 7.85 (s, 1H), 7.54 (dd, J = 5.4, 1.4 Hz, 1H), 7.46 (dd, J = 5.4, 3.4 Hz, 1H) ppm; ^{13}C NMR δ = 211.1, 177.1, 146.5, 139.0, 137.4, 133.4, 127.9, 127.8 ppm; HR-MS (APCI-MS, positive): m/z = 244.9212. Calcd for $\text{C}_8\text{H}_5\text{OS}_4^+$: 244.9218 $[M+H]^+$.

4.1.18 | Preparation of LiTMP

2,2,6,6-Tetramethylpiperidine (1.70 mL, 1.41 g, 10 mmol) was dissolved in THF (8.3 mL). $n\text{BuLi}$ (1.6 M in hexanes, 6.25 mL, 10 mmol) was added dropwise to the mixture at -40°C . The resulting mixture was stirred at the same temperature for 2 hr and 0°C for 1 hr to provide a hexanes/THF solution of LiTMP. Concentration of basic compounds was determined by titration using benzoic acid and 4-(phenylazo)diphenylamine to be 0.58 M.

4.1.19 | Preparation of $\text{TMPZnCl}\cdot\text{LiCl}$

ZnCl_2 (450 mg, 3.3 mmol) was suspended in THF (3.0 mL). LiTMP (0.58 M in hexanes/THF, 5.2 mL, 3.0 mmol) was added dropwise to the mixture at 0°C . The resulting mixture was stirred at the same temperature for 2 hr to provide a hexanes/THF solution of $\text{TMPZnCl}\cdot\text{LiCl}$. Concentration of basic compounds was determined by titration using benzoic acid and 4-(phenylazo)diphenylamine to be 0.35 M.

4.1.20 | Deuteration of 4-substituted-1,3-dithiole-2-thione with $\text{TMPZnCl}\cdot\text{LiCl}$

Deuteration of **3a** is representative. **3a** (67.3 mg, 0.30 mmol) was dissolved in THF (1.0 mL). $\text{TMPZnCl}\cdot\text{LiCl}$ (0.35 M in hexanes/THF, 1.02 mL, 0.36 mmol) was added dropwise to the mixture at 0°C . The resulting mixture was stirred at the same temperature for 1 hr. After an addition of D_2O (2 mL), the resulting mixture was stirred at 25°C for 1 hr. The mixture was poured into sat. NaHCO_3 aq. and organic compounds were extracted with CH_2Cl_2 (10 mL $\times$ 3). The combined organic layer was dried with anhydrous sodium sulfate, passed through activated alumina, and concentrated in vacuo. The yield and H/D ratio were determined by ^1H NMR with diphenylmethane as an internal standard to be 95% and 5/95, respectively.

4.1.21 | 4,5-Dipivaloyl-1,3-dithiole-2-thione (5)

Compound **4a** (65.5 mg, 0.30 mmol) was dissolved in THF (1.0 mL). $\text{TMPZnCl}\cdot\text{LiCl}$ (0.35 M in hexanes/THF,

1.02 mL, 0.36 mmol) was added dropwise to the mixture at 0°C . The resulting mixture was stirred at the same temperature for 1 hr. After additions of CuI (9.5 mg, 0.050 mmol) and pivaloyl chloride (74 μL , 72 mg, 0.60 mmol), the resulting mixture was stirred at 0°C for 3 hr and quenched with water (3 mL). The mixture was poured into sat. NaHCO_3 aq. (40 mL) and organic compounds were extracted with CH_2Cl_2 (10 mL $\times$ 3). The combined organic layer was dried over anhydrous sodium sulfate, passed through a pad of activated alumina, and concentrated in vacuo. Recrystallization from CH_2Cl_2 /hexane provided **5** (80.9 mg, 0.27 mmol, 89%) as a yellow solid. ^1H NMR δ = 1.26 (s, 18H) ppm; ^{13}C NMR δ = 207.8, 199.3, 144.0, 44.4, 26.5 ppm; HR-MS (APCI-MS, positive): m/z = 303.0543. Calcd for $\text{C}_{13}\text{H}_{18}\text{O}_2\text{S}_3^+$: 303.0542 $[M+H]^+$.

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